MYP 5 SCIENCES — CRITERION B: INQUIRING AND DESIGNING

Lesson 5 practice: analysing a toy-car investigation plan (task-specific interpretation)

The centre column is a paraphrased summary of the Year 5 achievement bands—not the official rubric text. Use the school-approved rubric in Toddle for formal assessment. This task is formative practice; the four sections A1–A4 are not four separate official grades.

| Band | Year 5 Criterion B — general progression (paraphrased) | Toy-car planning task — what evidence could look like |
|---|---|---|
| 0 | Does not yet demonstrate a standard described in the higher bands. | A1–A4: An investigation plan cannot be identified or meaningfully assessed. |
| 1–2 | Basic problem statement, simple prediction, named variables and a method that is difficult to follow. | A1: Mentions cars changing speed. A2: Predicts faster/slower with little explanation. A3: Names time and velocity with limited controls. A4: Gives a brief sequence but lacks a reliable measurement plan. |
| 3–4 | Outlines a testable problem, a reasoned prediction, variable choices and a safe outline method with apparatus. | A1: Identifies a fixed incline and what changes with time. A2: Predicts a pattern with a simple causal idea. A3: Names IV/DV and some controlled factors. A4: Provides safe apparatus and a rough timing/graphing method. |
| 5–6 | Describes the problem clearly, explains a relevant hypothesis, describes controls and sufficient measurements, and gives a complete safe method. | A1: Defines a measurable car-on-ramp relationship. A2: Uses force and acceleration reasoning. A3: Explains how time/velocity are measured and controls such as same ramp/car. A4: Gives marked positions, suitable repeats, recording procedure and usable data range. |
| 7–8 | Explains the scientific problem and prediction convincingly; justifies how variables and data will address it; specifies a coherent, repeatable and safe procedure. | A1: Gives precise, justified research question and release conditions. A2: Correctly links force, acceleration and expected velocity pattern. A3: Justifies important controls, sufficient repeated data and measurement choices. A4: Replicable method with ramp angle, marker range, repeats, handling of timing, units and safety. |

Strands to consider: i. Research question and explanation (A1); ii. hypothesis and scientific reasoning (A2); iii. variables and data collection (A3); iv. investigation design and method (A4).

Reference for full Year 5 wording: MYP Sciences, Criterion B, available in the school rubric and at https://www.rfs.edu.ps/uploads/files/MYP%20Years%204-5%20%20assessment%20criteria%201.pdf